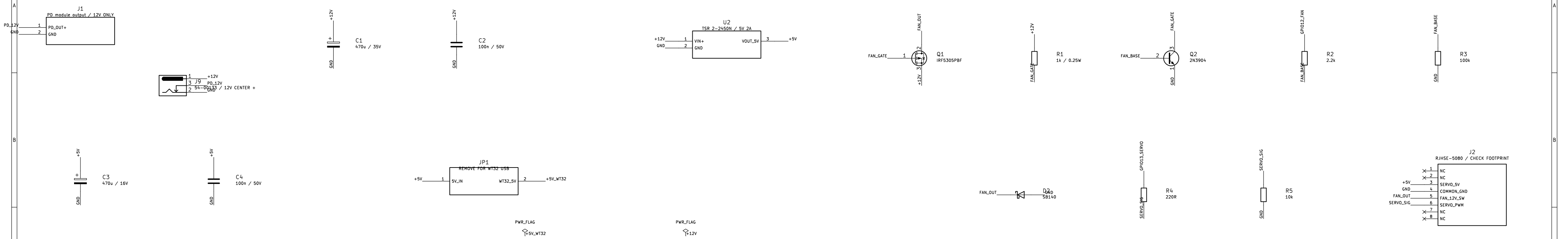


01 POWER – SWITCHED DUAL 12 V INPUT / FIXED 5 V BUCK

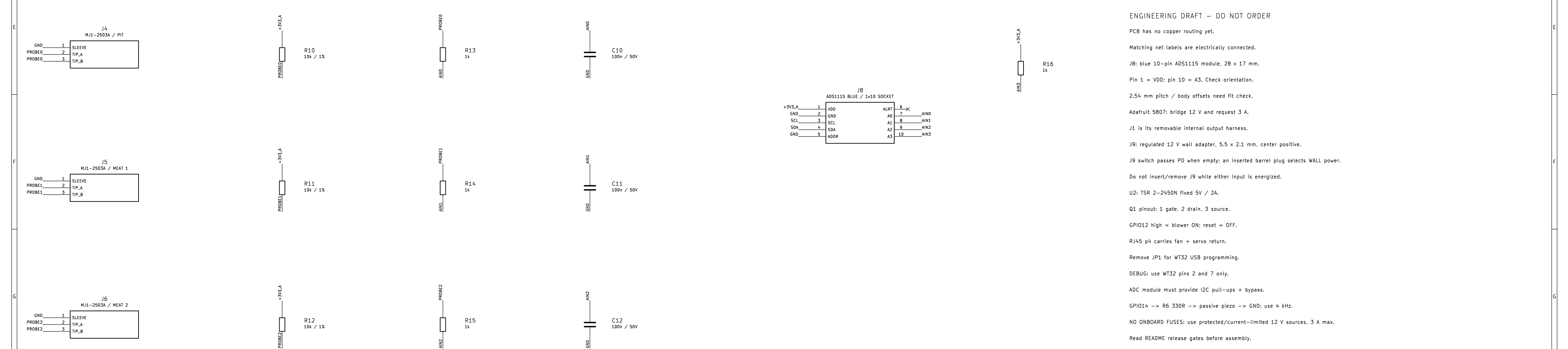
02 HIGH-SIDE BLOWER / SHARED-GROUND RJ45



03 WT32 HARNESSES / ANALOG POWER / DIRECT-DRIVE PIEZO



04 THREE NTC PROBES / FILTERS / SOCKETED ADC



ENGINEERING DRAFT – DO NOT ORDER

PCB has no copper routing yet.

Matching net labels are electrically connected.

J8: blue 10-pin ADS1115 module, 28 x 17 mm.

Pin 1 = VDD; pin 10 = A3. Check orientation.

2.54 mm pitch / body offsets need fit check.

Adafruit 5807: bridge 12 V and request 3 A.

J1 is its removable internal output harness.

J9: regulated 12 V wall adapter, 5.5 x 2.1 mm, center positive.

J9 switch passes PD when empty; an inserted barrel plug selects WALL power.

Do not insert/remove J9 while either input is energized.

U2: TSR 2–2450N fixed 5V / 2A.

Q1 pinout: 1 gate, 2 drain, 3 source.

GPIO12 high = blower ON; reset = OFF.

RJ45 p4 carries fan + servo return.

Remove JP1 for WT32 USB programming.

DEBUG: use WT32 pins 2 and 7 only.

ADC module must provide I2C pull-ups + bypass.

GPIO14 -> R6 330R -> passive piezo -> GND; use 4 kHz.

NO ONBOARD FUSES: use protected/current-limited 12 V sources, 3 A max.

Read README release gates before assembly.

NTC: $R = 10000 * raw / (raw_supply - raw)$. AIN3 measures excitation. Fix firmware before testing.

NOT FOR FABRICATION – MODULES, MOUNTING AND LOADS REQUIRE CONFIRMATION